



Checking and securing dividing a fraction by a whole number

Task A: Contexts for dividing fractions

1) A pizza is split into 12 equal pieces. A teacher takes 2 pieces and the remaining is given equally between 2 students. What fraction of the whole pizza does each student receive?

Removing 2 pieces means there are 10 pieces left

Dividing these 10 pieces by 2 means each person gets 5 pieces each

This means each person receives $\frac{5}{12}$

2) There are 14 crayons in a box. Two are lost and the rest are shared equally between a table of 4 students. What fraction of the full crayon box does each student receive?

Removing 2 crayons means there are 12 left

Dividing these 12 pieces by 4 means each person gets 3

This means each person receives $\frac{3}{14}$

3) A chocolate bar has 24 equal pieces. The teacher eats 6 pieces and shares the rest between 3 students. The teacher says, everyone has had the same fraction of the chocolate bar. Is the teacher correct?

Removing 6 pieces means there are 18 pieces left

Dividing these 18 pieces by 3 means each person gets 6

This means that they all received the same fraction, $\frac{6}{24}$ or $\frac{1}{4}$ of the chocolate bar.

4) A pizza is cut into 4 equal pieces. There are 6 people who need to share the pizza equally. How should the pizza be cut so everyone receives an equal piece? What fraction does each person receive?

Cut each $\frac{1}{4}$ into 3 pieces

This means there are 12 pieces

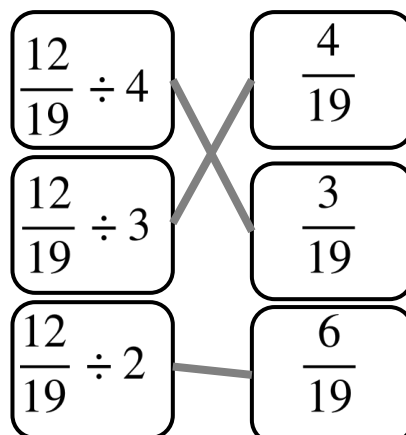
Each person then gets $\frac{2}{12}$

Which is the same as $\frac{1}{6}$



Task B: Dividing any fraction by a whole number

1) Match the questions and answers



2) Work out the answers to the following. You can draw bar models if needed.

a) $\frac{24}{25} \div 6 = \frac{4}{25}$

b) $\frac{21}{31} \div 7 = \frac{3}{31}$

c) $\frac{9}{11} \div 3 = \frac{3}{11}$

d) $\frac{10}{19} \div 5 = \frac{2}{19}$

3) Jacob has used equivalent fractions to work out his answer but has spilt ink all over his work. Can you identify what his missing numbers were?

a) $\frac{3}{2} \div 2 = \frac{3 \times 2}{2 \times 2} \div 2$
 $= \frac{6}{4} \div 2$
 $= \frac{3}{4}$

b) $\frac{5}{8} \div 4 = \frac{5 \times 4}{8 \times 4} \div 4$
 $= \frac{20}{32} \div 4$
 $= \frac{5}{32}$

4) Work out the answers to the following, giving your answer in its simplest form.

a) $\frac{5}{9} \div 4 = \frac{5 \times 4}{9 \times 4} \div 4 = \frac{20}{36} \div 4 = \frac{5}{36}$

b) $\frac{6}{11} \div 5 = \frac{6 \times 5}{11 \times 5} \div 5 = \frac{30}{55} \div 5 = \frac{6}{55}$

c) $\frac{10}{13} \div 4 = \frac{10 \times 2}{13 \times 2} \div 4 = \frac{20}{26} \div 4 = \frac{5}{26}$